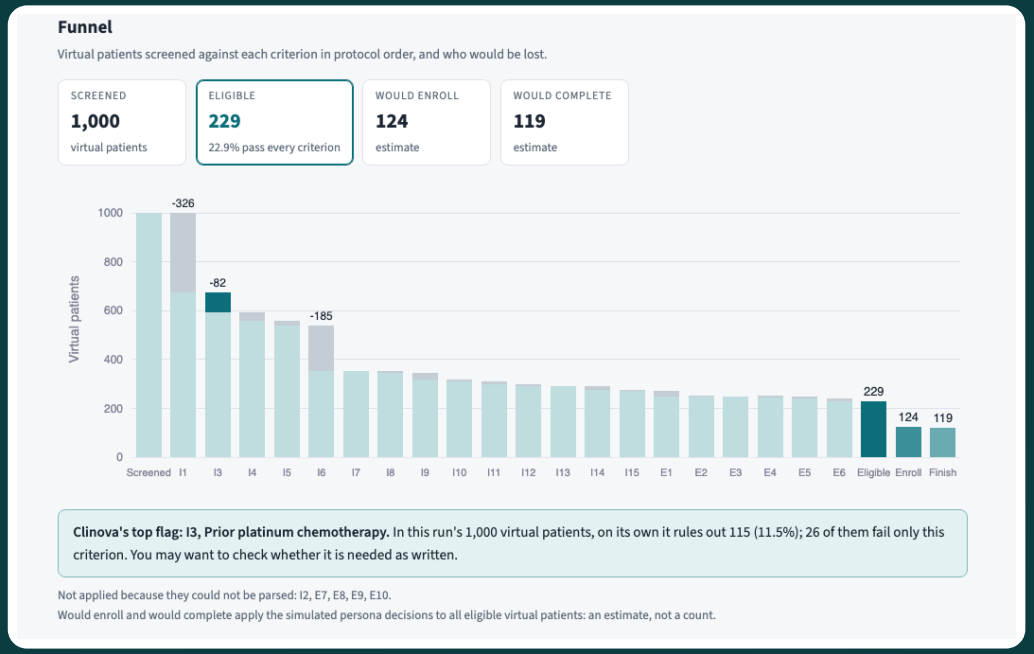
Clinova

Test-fly your NSCLC trial on thousands of virtual patients before enrolling a real one



Every prediction is graded against how real ClinicalTrials.gov trials turned out

https://github.com/RaghulKS/Clinova

THE PROBLEM

Most trial delays are written into the protocol before the first patient is screened

57%

of protocols need a substantial amendment

45% of those are avoidable

$535K

median cost of one Phase III amendment

$141K in Phase II, plus about 3 months

34.5%

of cancer-trial terminations: low accrual, the top reason

22.74% of 8,687 US cancer trials terminated

Clinova pulls the two levers the FDA cost study names: looser enrollment rules and simpler protocols

Sources: docs/ARCHITECTURE.md

WHY NOW

Public version history and LLM parsing make protocol risk checkable today

1

History is public

ClinicalTrials.gov keeps every registered version, so a widened rule is a checkable label

2

LLMs read eligibility

Gemini turns free-text criteria into rules that code can check on virtual patients

3

Design is the AI gap

Tufts CSDD 2024: full AI adoption is lowest in trial design and planning

Sources: docs/ARCHITECTURE.md

WHAT CLINOVA DOES

Clinova test-flies a protocol in five steps, from draft text to a graded risk

1

Read

Draft text or NCT ID to coded rules and visits

2

Simulate

Virtual NSCLC cohort, screened in code; persona agents decide

3

Calibrate

LightGBM on real trials, explained by SHAP

4

Optimize

Test edits; see trade-offs with ranges

5

Learn

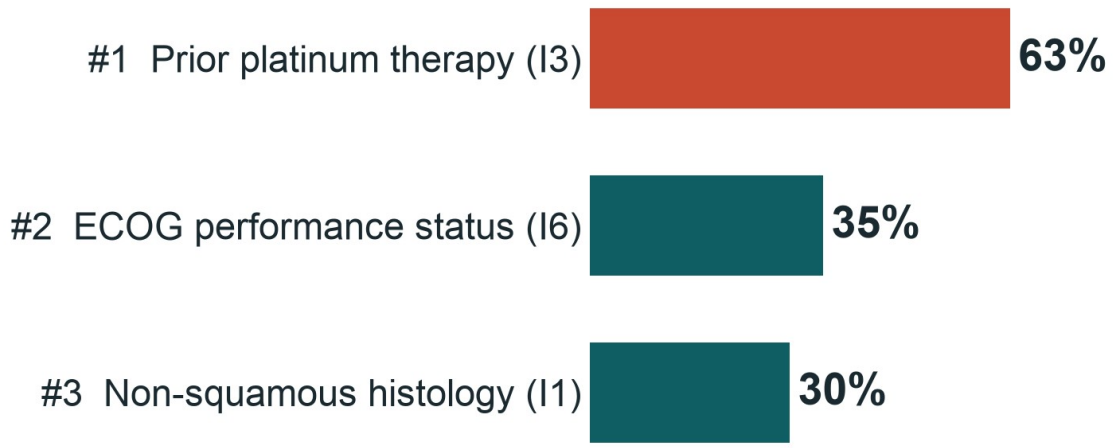
Log every prediction, score it when the trial ends

Sources: docs/ARCHITECTURE.md

BACKTEST ON A REAL TERMINATED TRIAL

Given only its first protocol, Clinova's flag #1 was the rule the sponsor later removed

Top 3 flags: share of virtual patients each rule excludes on its own



Coral: the rule the sponsor later removed

NCT03232892

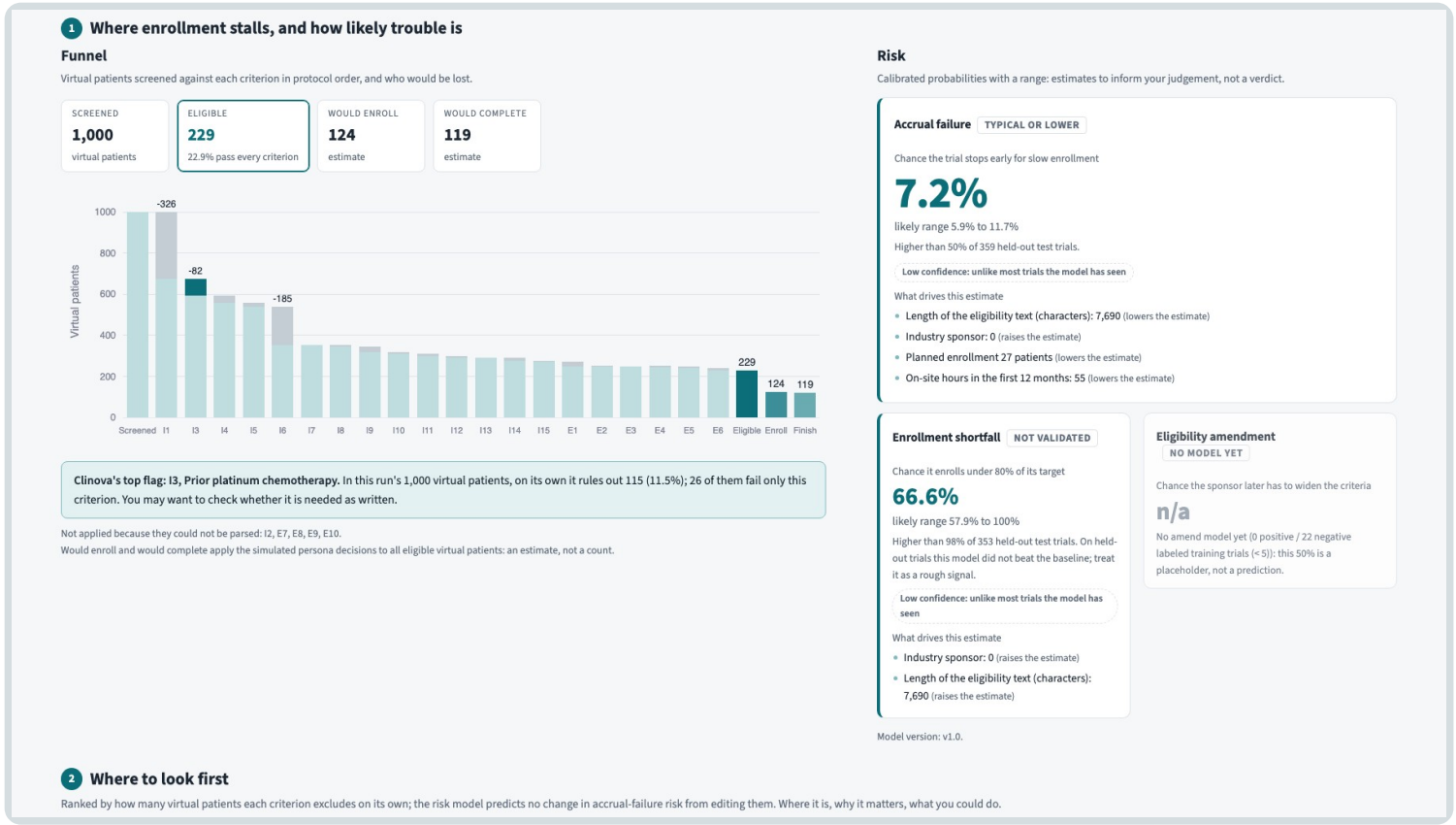
Terminated 2019-09-11: “Low Accrual”

- Flag #1: prior platinum therapy (I3), the rule the sponsor later removed; alone it excludes 63%
- Flags rank rules by how many virtual patients each excludes; the risk model does not read criteria
- Honest miss: predicted accrual-failure risk 7% (range 6% to 12%), the 50th percentile of test trials

Sources: docs/demo_trial.md, docs/demo_results/NCT03232892.json

LIVE DEMO

The app shows where enrollment stalls and how likely trouble is, in plain words



- Funnel: who is excluded by which rule, who enrolls, who completes
- Risk: probability with a range, its rank among test trials and plain-word drivers
- Labels say “not validated” or “no model yet” where the evidence is thin

Sources: docs/screens/03-funnel-risk.png

OPTIMIZER

The optimizer proposes three trade-offs with ranges; the author decides which, if any

Fastest enrollment

Widen histology rule (I1)
Lift ECOG limit (I6)
CrCl (I14): ≥ 50 to ≥ 20 mL/min

Enrollment speed	**+101%** +98% to +137%
Dropout	**+4.4 pts** +4.4 pts to +6.8 pts
Diversity index	**+0.039**

Fewest dropouts

Drop bilirubin rule (I11)
Drop histology rule (I1)
Drop cardiac exclusion (E6)

Enrollment speed	**+53%** +51% to +66%
Dropout	**-2.0 pts** -2.0 pts to +0.9 pts
Diversity index	**+0.019**

Balanced

ECOG (I6): ≤ 1 to ≤ 2
Drop histology rule (I1)
Drop CrCl (I14)

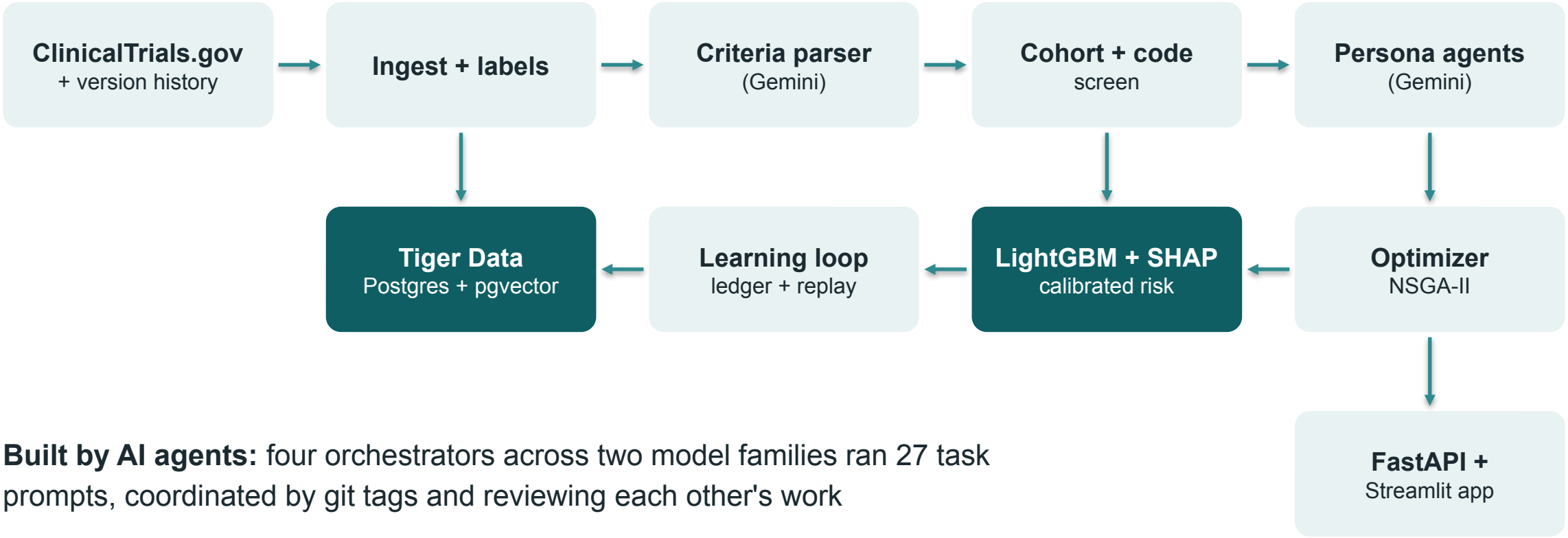
Enrollment speed	**+95%** +90% to +128%
Dropout	**+3.6 pts** +3.6 pts to +5.0 pts
Diversity index	**+0.037**

Ranges: 5 cohort seeds. No dollar value yet (no amendment labels). Authors lock rules that define the trial's population

Sources: docs/demo_results/NCT03232892_optimize.json

ARCHITECTURE

Small, tested modules around one Tiger Data store, built entirely by AI agents

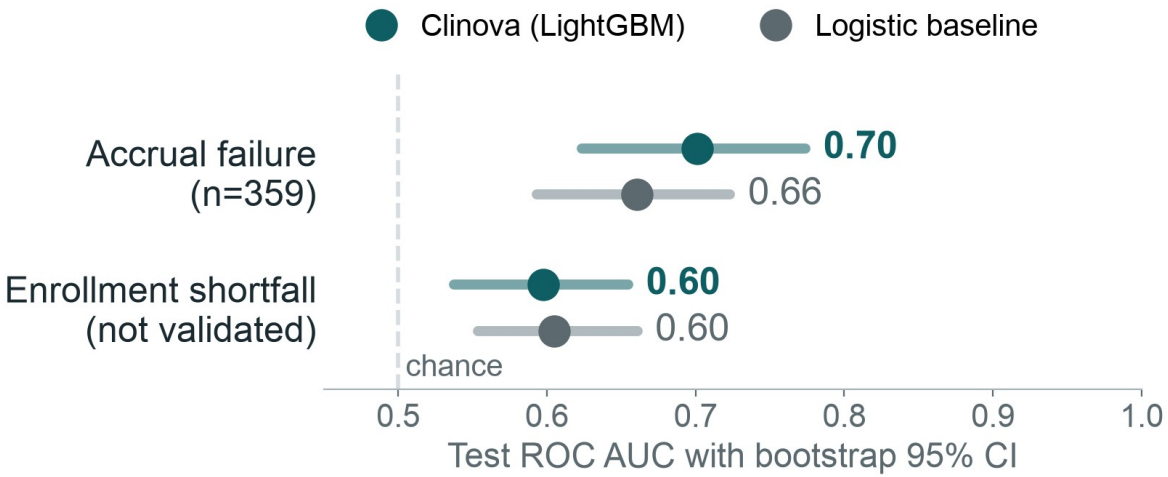


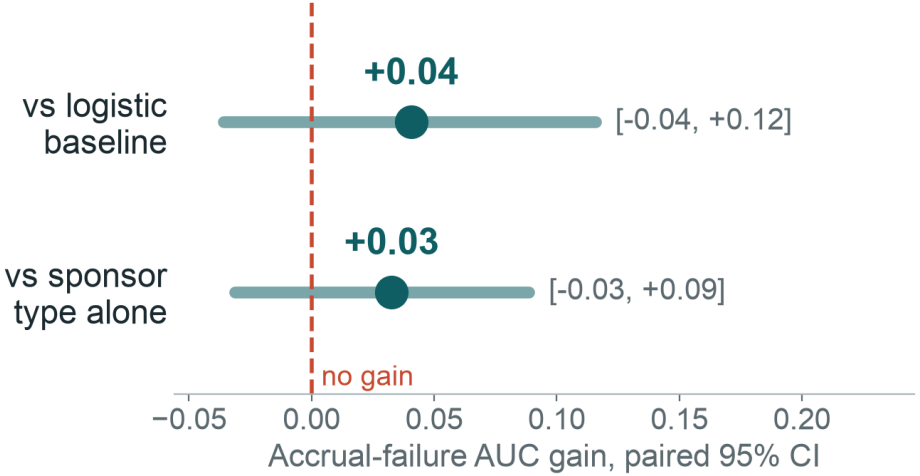
Built by AI agents: four orchestrators across two model families ran 27 task prompts, coordinated by git tags and reviewing each other's work

Sources: docs/ARCHITECTURE.md, orchestration/tasks/

VALIDATION ON TRIALS THE MODEL NEVER SAW

On 359 unseen trials, accrual-failure AUC is 0.70: above chance, not yet shown to beat a simple baseline



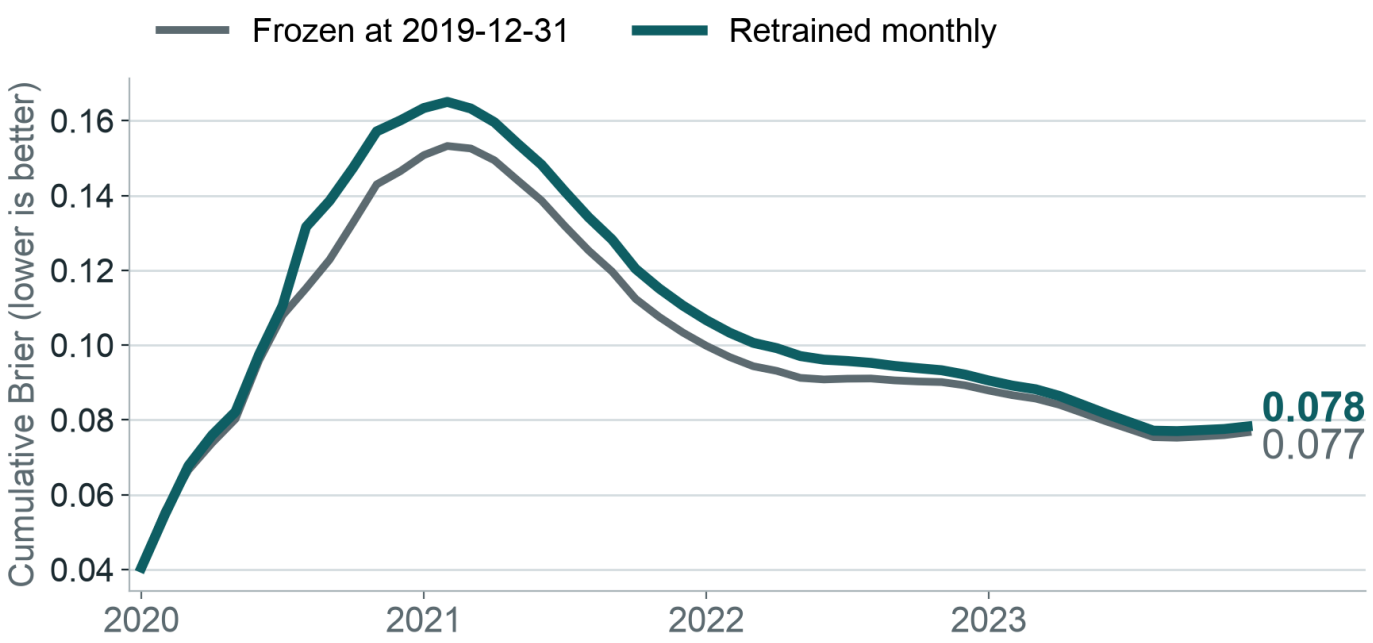


- Accrual failure: AUC 0.70 (95% CI 0.62–0.77) vs logistic baseline 0.66 (0.59–0.72)
- Paired AUC gain +0.04 (95% CI −0.04 to +0.12): the interval includes zero
- Brier 0.083 vs 0.086 for predicting the base rate; hit rate and simulation benefit not computable yet

Sources: docs/metrics.json, docs/metrics_detail.json

LEARNING LOOP

In a 2020–2023 replay, monthly retraining did not beat the frozen model (Brier 0.078 vs 0.077)



- Replay 2020-01 to 2023-12: score each month, then learn its outcomes
- Pooled Brier over 1,736 overlapping windows: retrained 0.078, frozen 0.077
- Every live prediction is logged in Tiger Data and scored when the trial ends

Sources: docs/replay.json

LIMITATIONS AND RESPONSIBLE USE

We show what the evidence can and cannot support

Not shown to beat simple baselines

Accrual AUC is above chance, but its paired gain over a logistic baseline includes zero

Amendment labels are missing

Version histories are not loaded yet, so hit rate and amendment risk are unmeasured

Personas are synthetic

Clinical traits come from published distributions; too few simulated trials to test their benefit

Scores are correlational

They rank protocols against similar trials; they do not show that a rule caused a failure

Clinova advises; the author decides. No real patient data. Every decision is logged.

Sources: docs/gate2_report.md, docs/metrics.json

ROADMAP

Next: more indications, site-level simulation and structured protocol input

1. Lymphoma, ophthalmology and immunology
2. Site-level enrollment simulation
3. Persona model tuned on expert feedback
4. Structured protocol input (ICH M11)
5. Full GxP validation package

Run it yourself

github.com/RaghulKS/Clinova

MIT license. docker compose up, then open the app; demo mode replays recorded Gemini API outputs.

Sources: docs/ARCHITECTURE.md